

INNOTRACK INTERNSHIP PROGRAM - 2026



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- **Project Name – STUDENT GRADE PREDICTOR**
- **Theme – MACHINE LEARNING IN EDUCATION**
- **VTU ID – VTU27739**



STUDENT GRADE PREDICTOR



INTRODUCTION:

Student performance prediction is an important application of Machine Learning in the education sector. The Student Grade Prediction System predicts a student's final grade using factors such as study time, failures, absences, and previous academic scores. The system helps educators and students analyze performance trends and make informed academic decisions. By utilizing data-driven prediction techniques, the project contributes to improved educational planning. By leveraging data-driven techniques, the system provides accurate predictions and personalized insights. This application demonstrates the practical use of Artificial Intelligence and Machine Learning in the education sector. The system analyzes important factors such as study time, failures, absences, and previous academic scores to estimate future performance. The project aims to assist students and educators in understanding academic trends and making informed decisions.

Idea:

- Develop an intelligent system to predict student academic performance.
- Utilize historical educational data to estimate final grades accurately.

Solution:

- Build a Machine Learning model that predicts final grades using student.
- Provide a Flask-based web interface for easy interaction and prediction generation.

Uniqueness:

- Predicts student academic performance using Machine Learning algorithms.
- Provides quick and automated grade prediction based on multiple academic factors.
- Combines data analysis, machine learning, and web development in a single application.
- Uses historical student performance data to generate meaningful insights.
- User-friendly web interface that allows easy interaction without technical knowledge.

TECHNICAL APPROACH

METHODOLOGY



Technologies to be Used:

- Frontend: HTML, CSS, JavaScript
- Backend: Python, Flask
- Machine Learning: Ridge Regression, Scikit-learn
- Database: SQLite
- Data Processing: Pandas, NumPy
- Methodology:**
- Dataset Collection
- Data Cleaning and Preprocessing
- Exploratory Data Analysis
- Feature Selection
- Model Training and saving

FEASIBILITY AND VIABILITY



FEASIBILITY

- Technical feasibility is achieved using Python and Flask frameworks.
- Machine Learning libraries provide efficient model development.
- Dataset availability enables effective model training.
- System requires minimal hardware resources.

VIABILITY:

- Provides real-world value in education.
- Helps identify at-risk students early.
- Supports data-driven academic decisions.
- Easy to use and maintain.
- Can be updated with new data.

CHALLENGES FACED

- Understanding and cleaning educational datasets.
- Handling missing or inconsistent data values.
- Selecting suitable Machine Learning algorithms.
- Improving prediction accuracy.
- Feature selection and correlation analysis.
- Integrating Machine Learning with Flask.
- Managing project deployment on cloud platforms.
- Testing application functionality across different scenarios.

FUTURE ENHANCEMENTS

- Integrate advanced Machine Learning algorithms.
- Increase prediction accuracy using larger datasets.
- Add student performance visualization dashboards.
- Implement user authentication and role management.
- Provide personalized learning recommendations.
- Support multi-institution academic analysis.
- Enable mobile-responsive application design.
- Integrate real-time educational analytics.

IMPACT AND BENEFITS

IMPACTS:

- Enables early identification of students who may require academic support.
- Promotes data-driven decision-making in educational institutions.
- Improves monitoring and analysis of student performance trends.
- Demonstrates the practical application of Machine Learning in education.
- Contributes to better academic planning and improved learning outcomes.

BENEFITS:

- Provides quick and accurate prediction of student grades.
- Helps teachers offer personalized guidance and support.
- Assists students in understanding and improving their academic performance.
- Reduces manual effort involved in performance evaluation.
- Offers a user-friendly web interface for easy accessibility and usage.

CONCLUSION

The Student Grade Predictor successfully demonstrates the application of Machine Learning in the education sector. By analyzing important academic factors, the system predicts student performance and provides valuable insights for educators and learners. The project combines data analysis, Machine Learning, and web development into a practical solution that supports academic decision-making. With future enhancements and wider adoption, the system has the potential to improve educational outcomes and promote intelligent learning environments. Making it a useful tool for supporting students in improving their academic performance. The Student Grade Prediction System was successfully developed and deployed, providing accurate grade predictions and personalized study recommendations .

THANK YOU